

1. Power

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ESP32 D1 R32 connections

5V pin -> +5V supply rail (red wire)

GND pin -> common ground (black wire)

3V3 pin -> sensor/DS18B20 only (orange wire)

Important

Never connect external 5V to the board while USB is also powering it.

All GND points must be common.

2. Phototransistor Sensor

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BPW40/BPW77 orientation

Check the exact datasheet for your part.

Collector -> sensor node / GPIO34

Emitter -> GND

Base (if present) -> not connected

Bias network

3V3 -> 1 Mohm resistor -> GPIO34 sensor node

Phototransistor collector -> same GPIO34 node

Phototransistor emitter -> GND

Firmware

Analog input: GPIO34

Alarm threshold is adjustable in Home Assistant.

3. Red Alarm LED

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LED pins

Long leg = anode (+)

Short leg = cathode (-)

Flat side of LED body usually marks cathode

Connection

ESP32 GPIO19 -> 330 ohm resistor

Resistor -> LED anode (long leg)

LED cathode (short leg) -> GND

Behavior

The red LED mirrors the boiler's alarm LED.

4. Green Status LED

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LED pins

Long leg = anode (+)

Short leg = cathode (-)

Flat side of LED body usually marks cathode

Connection

ESP32 GPIO21 -> 330 ohm resistor

Resistor -> LED anode (long leg)

LED cathode (short leg) -> GND

Behavior

The green LED remains on while the firmware is running.

5. 2N7000 and Active Buzzer

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2N7000 pins

Verify Gate/Drain/Source against the datasheet for your exact transistor.

Gate -> GPIO18 through 220 ohm

Gate -> GND through 100 kohm

Source -> GND

Drain -> buzzer negative (-)

Buzzer

Buzzer positive (+) -> +5V

Buzzer negative (-) -> MOSFET drain

Use an active 5V buzzer.

6. Mute Button

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Pushbutton pins

Use one pin from each internally connected side.

GPIO22 -> one side of button

Opposite side -> GND

Firmware

GPIO22 uses INPUT_PULLUP.

Pressed = LOW.

No external pull-up resistor is required.

7. DS18B20 Temperature Sensor

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Three-wire connection

VDD -> ESP32 3V3

GND -> ESP32 GND

DATA -> ESP32 GPIO23

Pull-up resistor

4.7 kohm between DATA and 3V3

Common waterproof-probe colors

Red = VDD

Black = GND

Yellow/White = DATA

Always verify the vendor datasheet.